



Exposure to microplastics leads to a defective ovarian function and change in cytoskeleton protein expression in rat

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Abstract

Microplastics (MPs) are ubiquitous environmental contaminants; through their physicochemical properties, they can have potentially negative effects on the environment as well as on animal and human health. Studies addressing the toxicity of MPs on mammalian female reproduction are almost absent. Thus, the main objective of the present study was to assess the impact of oral exposure, during four estrous cycles, of 5 µm polystyrene-type microplastics (PS-MPs) on ovarian function in rats. Particles of PS-MPs were detected in the duodenum and, for the first time, in the different compartments of the ovarian tissue. The toxicity of accumulated PS-MPs was manifested by the reduced relative ovarian weights, by the alteration in the folliculogenesis and in the estrous cycle duration, and by the reduced serum concentration of estradiol. The defective ovarian function following PS-MP treatment might be due to the induction of oxidative stress, which has been proved by an increased malondialdehyde (MDA) concentration and an increased superoxide dismutase (SOD) and catalase (CAT) activities as well as a decreased protein sulfhydryl (PSH) level in the rat ovary. Importantly, by immunofluorescence and RT-PCR, we demonstrated a significant decrease in the expression of cytoskeletal proteins: α -tubulin and disheveled-associated activator of morphogenesis (DAAM-1) in the ovary of rats exposed to PS-MPs at proteomic and transcriptomic levels. Our results uncovered, for the first time, the distribution and accumulation of PS-MPs across rat ovary, revealed a significant alteration in some biomarkers of the ovarian function, and highlighted the possible involvement of MP-induced disturbance of cytoskeleton in these adverse effects.

Keywords Polystyrene-microplastics · Ovary · Oxidative stress · Cytoskeleton · Rat

Introduction

Microplastics (MPs), particles with a diameter less than 5 mm, have been increasingly recognized as a global environmental threat and health hazard to animal and human populations (Galloway and Lewis 2016). Multiple studies have documented that MPs are ingested by various aquatic

organisms (bivalves, zooplankton, copepods, fish), accumulated in specific tissues (gill, gut, digestive gland, circulatory system) (Jeong et al. 2016; Wright et al. 2013; Zitouni et al. 2020), and are transported up through the food chains (Carbery et al. 2018). In these animals, tissue accumulation of MPs may induce various adverse effects, such as the reduction of feeding activity, inhibition of growth and development, energy deficiency (Lu et al., 2016), immune responses (Park et al. 2020), oxidative stress (Wan et al. 2019), metabolic disorder (Missawi et al. 2020; Zitouni et al. 2020), and genotoxicity (Zitouni et al. 2020). Recent studies have indicated that exposure of aquatic organisms to MPs induced defective female reproductive ability (Sussarellu et al. 2016; Wang et al. 2016; Felten et al. 2020).

While most researches on the toxic effects of MPs have focused on aquatic organisms, studies documenting the potential health risk and tissue accumulation of MPs in mammals are lacking (Deng et al. 2017; Amereh et al.

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2020). Hence, studies on tissue accumulation of MPs and their reproductive toxicity in mammalian models would be indispensable for the risk assessment of MPs in human health, especially female fertility. Polystyrene microplastics (PS-MPs) are one of the most commonly produced and used plastics; they have become model particles for studying biological effects and uptake in animal organisms. Recently, fluorescent PS-MPs (FPS-MPs) have been used to study the distribution, accumulation, and tissue-specific health risks of PS-MPs in mice (Deng et al. 2017; Lu et al. 2018a). The obtained results indicate that 5 μm FPS-MPs accumulate in the liver, kidneys, and intestines of exposed mice and result in multiple energy and lipid metabolism disorders and oxidative stress as well as a significant change in the intestinal microbiota (Deng et al. 2017; Lu et al. 2018a).

A decrease in energy allocation and a disturbance of oxidative status can play a major role in the impairment of the reproductive function (Sussarellu et al. 2016; Prokic et al. 2019). Amereh et al. (2020) presented the first appraisal in rat of the impact of exposure on reproductive toxicity. Using fluorescently labeled polystyrene (PS) nanoplastics, authors observed serious histological lesions and semen biomarker alterations in testis tissue (Amereh et al. 2020). In the two studies available in the literature to date, evaluating the effect of rat exposure to MPs on ovarian function (An et al. 2021; Hou et al. 2021), it has been shown that 0.5 μm PS-MPs can enter granulosa cells (GC), following in vitro exposure, causing reduction in the number of growing follicles and inducing oxidative stress as well as apoptosis of GC. Oxidative stress, due to its deleterious effects on proteins, is postulated to be the one of the important mechanistic pathways in female infertility (Gupta et al. 2014). It is well established that oxidative stress negatively influences ovarian function by directly disrupting oocyte maturation (Zhang et al. 2016) and/or indirectly by affecting the integrity of the cytoskeleton of follicular cells. In fact, cytoskeleton assembly is crucial for oocyte maturation, and the disorder of microtubule dynamics will cause spindle formation defects, which further induce cell division failure (Hu et al. 2018).

The main objective of the present study was to assess the impact of oral exposure, during four estrous cycles, of 5 μm PS-MPs on ovarian function in rats. A particular interest has been attributed to the study of the accumulation of PS-MPs in the ovary and its effect on some parameters of reproductive function, namely estrous cyclicity, folliculogenesis, and circulating estradiol (E_2) level. It is also important to know if the accumulated PS-MPs in the ovarian tissue affect the oxidative status and the expression of two proteins involved in the integrity of the cytoskeleton: α -tubulin as a structural protein, and disheveled-associated activator of morphogenesis (DAAM-1) as a regulatory protein.

Material and methods

Polystyrene microplastic particles

Two types of PS-MPs were used in this study and were obtained from microparticles GmbH (Germany, Lots: PS-FluoGreen-Fi199 and PS/Q-R-KM491). One type was fluorescent particles of polystyrene (FPS-MPs) with excitation and emission wavelengths of 502 and 518 nm, respectively. This type of particle was used to evaluate the accumulation and distribution of MPs in the duodenum and ovary of rats. The second type was pristine particles (PPS-MPs), which were used for toxicological tests (Deng et al. 2017). These two types were spherical particles with a diameter of 5 μm . The size of the particles was measured with non-invasive backscatter optics. The ENM ζ -potential was also measured via Zetasizer Nano ZS. Particle size was independently monitored by our partner in CNRS, ICMPE, University of Creteil, Paris, France. An in-house assessment of the polystyrene spheres revealed an average particle agglomerate size of $4.98 \mu\text{m} \pm 0.15$ and a ζ -potential of -0.0934 ± 0.172 . In the present study, specific PS microspheres were used instead of heterogeneous and more realistic environmental MPs in order to limit the key factors that may affect uptake and translocation of micro-scaled plastics. In addition, PS-MPs are also widely applied in bioassays examining biological interactions and toxicities in organisms (Yang et al. 2019).

Animals and experimental design

Twenty-one female adult Wistar rats *Rattus norvegicus*, with a regular estrous cycle, weighing 196 ± 11 g were used in this study. Each animal was housed in individual cage at 23 ± 1 °C and exposed to a 12-h light–dark cycle. All animals had access to a standard rodent laboratory, a fixed formulation fortified with vitamins (750 μg riboflavin, 250 μg acide folique, 19 μg acide ascorbique, 5 μg cyanocobalamine, and 40 μg biotine per 100 g) and minerals (173 mg NaCl, 343 mg $\text{NaHPO}_4\text{H}_2\text{O}$, 945 mg K_2HPO_4 , 600 mg MgCl_2 , 0.3 mg CuSO_4 , 0.3 mg CoCl_2 , 0.3 mg ZnCO_3 , 4.0 mg MnSO_4 per 100 g) and drinking water. The animals were housed according to the Ethics and Confidentiality Committee (EEC) 609/86 Directives regulating the welfare of experimental animals and the experiments were approved by the local ethics committee of Institute of Biotechnology, University of Monastir, (Ref: CER-SVS/ISBM022/2020). The rats were weighted every 5 days to control the weight progression during the duration of MP exposure. Estrous cyclicity was examined three times per day by vaginal smear preparation (Viguié

et al. 2018). Animals in the estrus phase were selected and randomly divided into two groups; a control group ($n = 7$) receiving, by oral gavage, PS-MP-free ultrapure water and a group ($n = 14$) received PS-MPs daily by oral gavage (0.1 mg/day, which corresponds to 1.5×10^6 particles/day) during four estrus cycles. Concerning the treated group, 7 animals received a fluorescent form of PS-MPs (FPS-MPs) and were used for the study of the tissue accumulation of MPs and the other animals received PPS-MPs and were used for the study of all the other parameters. It should be noted that the two forms of PS-MPs have the same physic-chemical properties and only differ by the emission of fluorescence. Data related to the concentrations of PS-MPs in natural environments or food chains are not available. Thus, the concentration used in our study may not correspond to environmentally relevant concentrations; it was selected in the aim of providing insights into effect of PS-MPs on physiological function before evaluating the risk at environmental concentrations. The chosen concentration of 0.1 mg/day was also used in previous studies (Deng et al. 2017), which caused accumulation and some adverse effects in tissues of mice. The period of four estrus cycles corresponds to 24–26 days and seems sufficient to study the tissue accumulation of PS-MPs and its possible effect on the ovarian function. This period is comparable to that used in order to study the accumulation and reproductive toxicity of PS-MPs in mice (Deng et al. 2017, 2021; Amereh et al. 2020; Jin et al. 2021). In this experiment, PS-MP particles were dispersed in ultrapure water and were treated with ultrasonic vibration for 20 min.

Sample collection

In the estrus phase of the 5th estrous cycle, the final body weights of each rat, which present an initial body weight of 196 ± 11 g, were registered and the animals were anaesthetized by 4% chloral hydrate (i.p. 10 mL/Kg), and then sacrificed. Before the animal's death, blood was taken by cardiac puncture, collected in sterile tubes, and then centrifuged at a speed of 5000 rpm/min for 10 min at $+4$ °C. The collected serum was stored at -80 °C for estradiol assay. Right ovaries were stored at -80 °C for biochemical analysis or placed in RNA later and stored at -20 °C until RNA extraction and molecular analysis, while left ovaries were fixed in 10% formalin buffer and processed for histology and immunofluorescence studies. The ovaries were weighed by the analytical balance to determine their relative weights per 100 g of the body weight of the rats.

Histological and immunofluorescence examination

Localization of fluorescent PS-MPs in the duodenum and the ovary

The ovary and duodenum of the group exposed to FPS-MPs were fixed in 10% formalin, embedded in paraffin wax, sectioned at 4 μ m thickness, and stained with hematoxylin and eosin (H&E) for microscopic examination. In order to observe the presence of FPS-MPs in tissues, one bright-field image was acquired by microscopy first and then a dark-field image of the same slide was acquired by epifluorescence microscopy (Zeiss, Axiostar plus, Germany) (Deng et al. 2017). Finally, the two sets of images were merged using ImageJ (National Institutes of Health, Bethesda, MD, USA) (Deng et al. 2017). The quantification of the mean fluorescence intensity was calculated using ImageJ (software v.2.0.0). For each rat, image acquisition was done from 3 different sections and three areas, of each section was taken into consideration for quantification of the fluorescence intensity.

Histological analysis

Ovaries were fixed in 10% formalin for one night. After that, they were dehydrated in rising alcohol levels before paraffin was added. Sections of the ovaries (4 μ m) were obtained, stained with H&E, and examined under an optical microscope (Zeiss). Folliculogenesis was analyzed by quantifying follicles in each stage of folliculogenesis: primordial, primary, secondary, antral, and atretic follicles. Additionally, cysts and corpus luteum were identified. The follicle count was made, at low magnification, over the entire section, making three sections for each individual at different levels. The follicles were classified according to Towell-established criteria (Hirshfield 1987).

Immunofluorescence analysis

For α -tubulin and DAAM-1 localization, 4 μ m ovary sections were deparaffined, rehydrated, and processed as described by Pariante et al (2016). Antigen retrieval was performed by pressure-cooking slides for 3 min in 0.01 M citrate buffer (pH 6.0). Non-specific binding sites were blocked with an appropriate normal serum diluted 1:5 in phosphate-buffered saline (PBS) containing 5% (w/v) bovine serum albumin (BSA) before the addition of anti- α -tubulin (ab 52,866) or anti-DAAM-1 (ab221571) antibody diluted 1:100, for overnight incubation at 4 °C. After washing in PBS, slides were incubated for 1 h with the appropriate secondary antibody goat anti-rabbit (Alexa fluor 488) diluted 1:500 in the blocking solution. The images were captured with the epifluorescence microscopy (Zeiss, Axiostar plus).

α -tubulin and DAAM-1 protein levels were estimated by the quantification of the green fluorescence intensity using ImageJ (software v.2.0.0). For each rat, image acquisition was done from 3 different sections and three areas, of each section was taken into consideration for quantification of the fluorescence intensity.

Biochemical analysis

Oxidative stress markers

The enzymatic (activities of antioxidant enzyme superoxide dismutase (SOD) and catalase (CAT) and non-enzymatic markers of oxidative stress) levels of malondialdehyde (MDA) and protein sulfhydryl (PSH) were assessed in the ovarian homogenates. The ovaries were homogenized in phosphate buffer (0.1 M; pH 7.4) at a ratio of 1:1 w/v at 4 °C. Then, the homogenates obtained were centrifuged at 12,000 g for 30 min at 4 °C, and the supernatants were recovered for subsequent analyses.

Total protein content in the homogenate was measured following the Bradford method (Kruger1994). The activity of SOD was assessed by the method of MarklundS (1974). This method is based on the inhibitory effect exerted by SOD against the pyrogallol autoxidation. The maximal absorbance and the absorbance of samples were determined at 420 nm in the presence of pyrogallol. The activity of CAT in the samples was measured at 240 nm by the method of Aebi (1984), using hydrogen peroxide as a substrate. SOD and CAT activities were expressed as units per milligram protein. Lipid peroxidation product was assayed by thio-barbituric acid (TBA)-reactive substance method modified from that of Yagi (1976). This method consists in reacting the aldehyde functions released at 95 °C by acid hydrolysis (perchloric acid) to form an MDA-(TBA) complex (reddish color) quantifiable at 535 nm by spectrophotometry.

PSH concentration was determined by the method of Sedlack and Lindsay (1968) by subtracting the non-PSH content from the total sulfhydryl content by 5% tri-chloroacetic (TCA). The absorbance was measured at 412 nm against sample-free and 5,5-dithio-bis-(2-nitrobenzoic acid)

(DTNB)-free blanks. The oxidative damage of proteins is marked by the decrease in the levels of PSH.

Estradiol assay

Serum levels of E₂ were evaluated using commercially available enzyme-linked immunoassay kits (BioVendor) according to the manufacturer's protocol. The assay was measured at an absorbance of 450 nm.

Molecular analysis

Total RNA was extracted from ovarian samples using PRImezol reagent (AN1102). The RNA purity was checked by optical density (OD260/OD280 absorption ratio > 1.8) and agarose gel electrophoresis 1.5%. A total amount of 1.5–2 µg of total RNA was reverse transcribed in a 20-µL reaction mixture using random hexamer primers (Roche), 0.5 mM dNTPs (Roche), and 19 M-MuLV RT buffer as described by (Banni et al. 2009). Briefly, the RNA was denatured by heating for 5 min at 70 °C, cooled on ice, and incubated with a reverse transcriptase reaction mixture. For reverse transcription, tubes were incubated at 42 °C for 60 min, followed by rapid cooling. The volume of the RT mixture was raised to 100 µL with nuclease-free distilled water, and 4 µL was used for the amplification of the gene targets. The messenger RNA (mRNA) abundances of the target genes were evaluated in the presence of 19 QuantiTectSybr Green PCR Master Mix (Qiagen). Primer pairs (Table 1) were designed using Beacon Designer v3.0 (Premier Biosoft International, Inc.). The amplifications were carried out for 40 cycles, with denaturation for 10 s at 95 °C, annealing for 20 s at 60 °C, and extension for 30 s at 72 °C (Banni et al. 2011). The relative expression of the genes of a sample was determined according to the formula of Livak and Schmittgen (2001): $2^{-\Delta\Delta Ct}$.

Statistical analysis

All data were shown as mean $\pm$ standard error (SE). Differences were considered significant when $p < 0.05$. Statistical

Table 1 qRT-PCR primers and accession numbers of the corresponding sequences used

Genes	Accession number	Primer sequences (5'–3')	Amplification size (bp)
DAAM-1	190,916-006_G4	F- GCCATGTTTGCACTTCCAGC	20
	190,916-006_G5	R- TCAGAATGAGCCAGGACATG	
α -tubulin	190,916-006_G6	F-CTCGCCTAGATCACAAGTTT	20
	190,916-006_G7	R-TATGAAAGCACACATTGCCA	
β -actin	190,916-006_G8	F-CTGGGAGTGGTTTGAGGTGT	20
	190,916-006_G9	R-CTATGCAGGTGGGAGGATGT	

analysis was completed by one-way ANOVA followed by Tuckey's post hoc *t*-test, and paired test where appropriate with Prism 5.0, GraphPad software.

Results

Accumulation and distribution of FPS-MPs in rat tissues

Figure 1A shows the accumulation of FPS-MPs in the duodenum and the ovary. These particles were clearly visible as distinct fluorescent points in the duodenum (Fig. 1A b and A c) and the ovary (Fig. 1A e and A f) of exposed rats; however, these particles were absent in the duodenum (Fig. 1A a) and the ovary (Fig. 1A d) of the control group. The results presented in Fig. 1B indicate that the FPS-MP accumulation is significantly higher ($p < 0.01$) in the duodenum than in the ovary. As shown in Fig. 1C, the FPS-MPs occupy a space in the ovary that differs from one structure to another. In fact, the corpus luteum and the atretic follicles show a significantly higher (respectively $p < 0.101$ and $p < 0.05$) percentage of accumulation of FPS-MPs than the other follicles and stroma.

Body weight gain and relative ovary weight.

As shown in Fig. 2A, a significant decrease ($p < 0.05$) in body weight gain was noted at the end of the experiment in PPS-MP-exposed animals. A significant decrease ($p < 0.05$) in the relative weight of ovaries was also noted in the PPS-MP-treated group compared with controls (Fig. 2B).

Effect of PS-MPs on some parameters of ovarian function

Effect on estrous cyclicity

In order to evaluate the effect of PPS-MP exposure on the duration of the estrous cycle, vaginal smear was determined three times per day. Our results showed that the PPS-MP exposure significantly decreased the duration of the fourth estrous cycle ($p < 0.01$) compared to the control group (Fig. 3A). Moreover, the study of PPS-MP exposure effect on each phase of the estrous cycle showed that these particles caused a very significant decrease ($p < 0.01$) in the duration of the metestrus phase compared to control (Fig. 3B).

Fig. 1 **A** Accumulation of fluorescent polystyrene microplastic particles (FPS-MPs) in rat tissues after exposure for 4 estrus cycles in the duodenum (A b and A c) and in the ovary (A e and A f) compared to the control duodenum (A a) and control ovary (A d) ($n = 7$ animals in each group). Overlay (background: hematoxylin–eosin) (A. a, b, d, and e: $Gr \times 10$; c and f: $Gr \times 40$) (scale bar = 50 μ m). **B** Histogram showing the quantification of fluorescence signal intensity in the ovary and the duodenum using ImageJ. a*: $p < 0.01$ when compared to duodenum. **C** Histogram showing the quantification of fluorescence signal intensity in the different ovarian structures: follicles (F), corpus luteum (CL), atretic follicle (AF), and stroma (S) in the treated group using ImageJ. a*: $p < 0.01$ when compared to the structure F; a: $p < 0.05$ when compared to the structure F; b*: $p < 0.01$ when compared to the structure CL; c*: $p < 0.01$ when compared to the structure AF

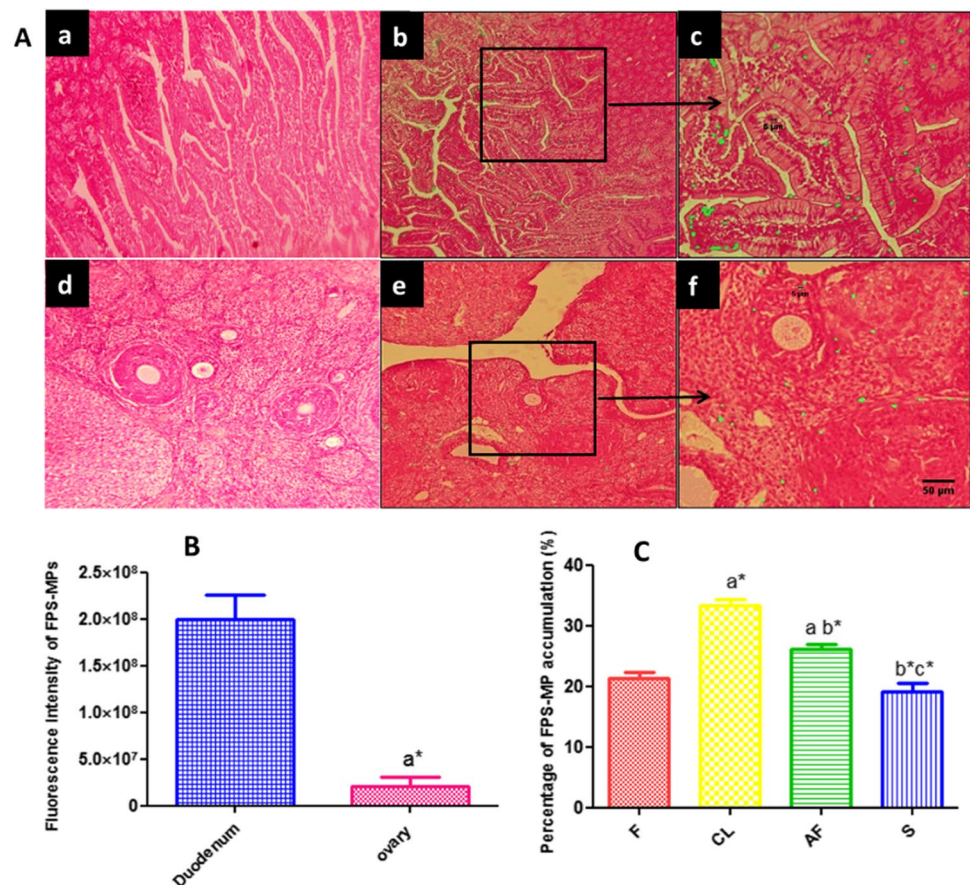
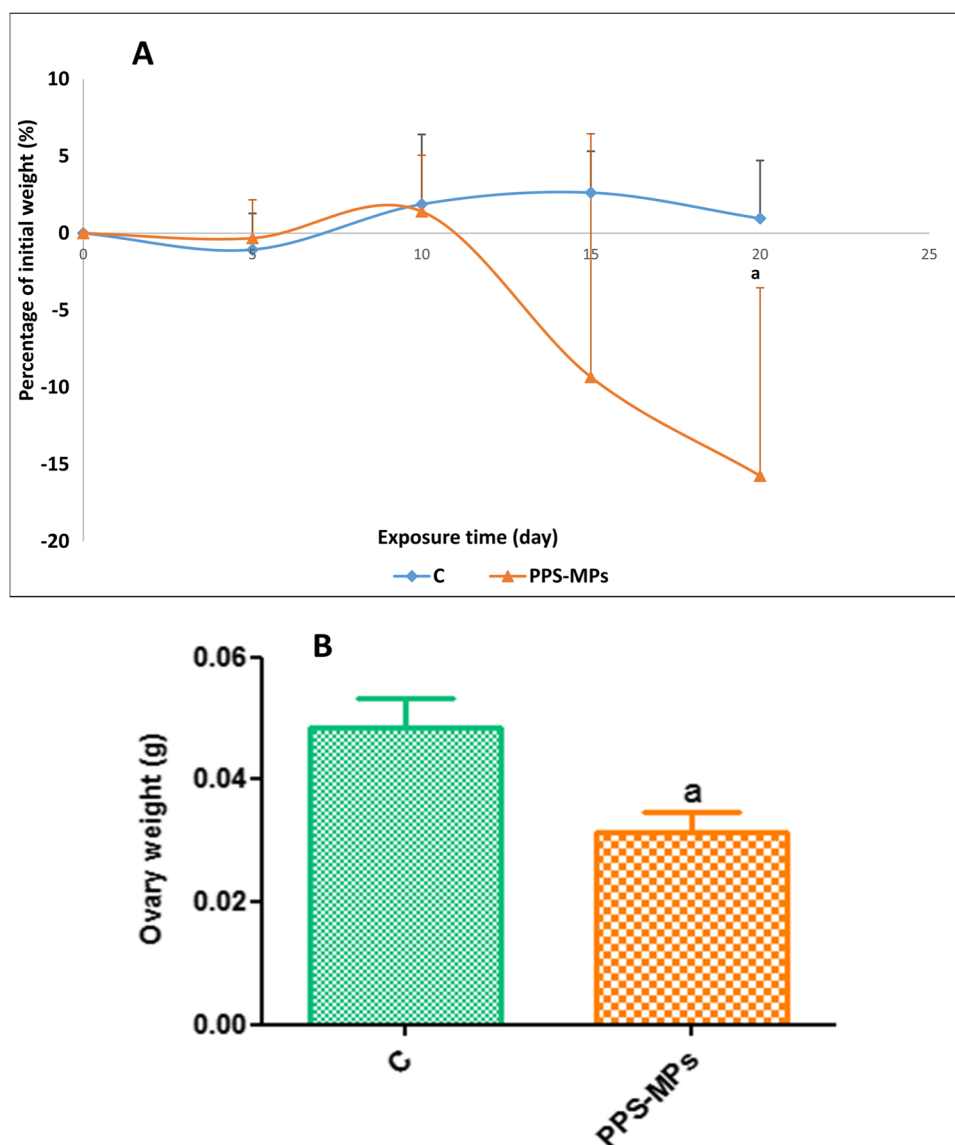


Fig. 2 Effect of pristine polystyrene microplastic particle (PPS-MP) exposure on the bodyweight gain (expressed as % of the initial weight) (A) and on the ovary relative weight (g/100 g body weight) (B) ($n=7$ animals in each group). a: $p<0.05$ when compared to control animals (C)



Effect on the histological structure of the ovaries and on folliculogenesis

The representative photomicrographs of ovaries are shown in Fig. 4A. The control group demonstrated the normal structure of rat ovary showing the different stages of folliculogenesis (Fig. 4A a). The ovaries of rats treated with PPS-MPs showed several alterations such as tissue hemorrhage, cell vacuolization, and granulosa cell degeneration (Fig. 4A: b, c, d, e, and f).

Moreover, the PPS-MP-exposed group showed an evident disruption of folliculogenesis marked by a very significant decrease ($p<0.01$) in the number of tertiary follicles and corpora lutea and a significant increase ($p<0.05$) in the number of primary and atretic follicles and cysts compared to the control group (Fig. 4B).

Effect on serum concentration of E_2

As shown in Table 2, the serum E_2 concentrations, determined in the estrus phase of the fifth cycle, decreased significantly ($p<0.05$) after exposure to PPS-MPs when compared to the control group.

Effect on selected markers of oxidative stress

The results in Table 2 show that PS-MP exposure caused a significant increase ($p<0.05$) in CAT activity and MDA levels as well as a non-significant increase in SOD activity. Besides, the exposure to PS-MPs resulted in a significant reduction ($p<0.05$) in PSH levels compared to control.

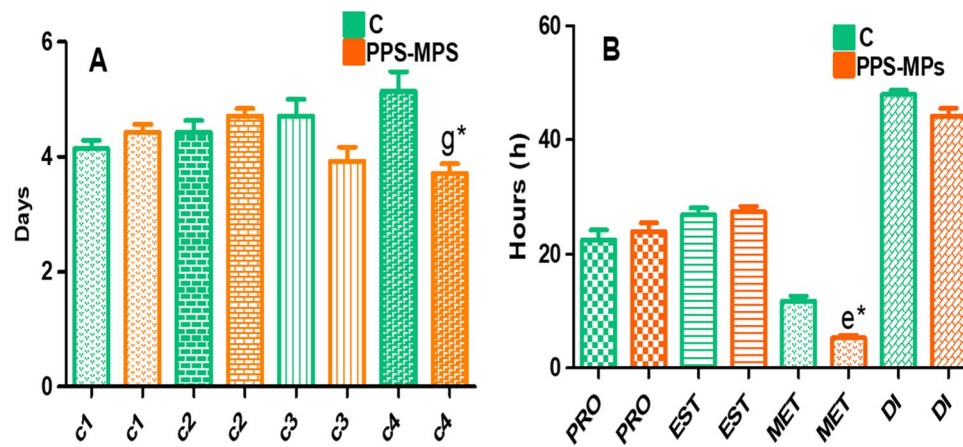


Fig. 3 Effect of pristine polystyrene microplastic particle (PPS-MP) exposure on the duration of four estrous cycles of rats (**A**) and on the duration of different phases of estrous cycle (**B**). Groups were identified as control (untreated) animals (C) and PS-MP-treated group ($n=7$ animals in each group) (*c1*, the first estral cycle; *c2*, the sec-

ond estral cycle; *c3*, the third estral cycle; *c4*, the fourth estral cycle; *PRO*, proestrus phase; *EST*, estrus phase; *MET*, metestrus phase; *DI*, diestrus phase). g^* : $p < 0.01$ when compared to the fourth cycle of controls animals (**A**); e^* : $p < 0.01$ when compared to the metaestrus phase of controls animals

Fig. 4 **A** Histological structure of ovaries from control rats (**A a**) and rats treated with pristine polystyrene microplastic particles (PPS-MPs) (**A b-c-d-e-f**) (H&E staining) ($Gr: \times 40$). **B** Histogram showing the effect of PPS-MP exposure on folliculogenesis ($n=7$ animals in each group) *PrF*, primordial follicle; *PF*, primary follicle, *SF*, secondary follicle, *CL*, corpus luteum; *AF*, atretic follicle; *CF*, cystic follicle; *GD*, granulosa degeneration; *V*, vacuolation; *H*, hemorrhage; *TF*, tertiary follicle; *GF*, graafian follicle. *a*: $p < 0.05$ when compared to the number of primary follicles (PF) of control group (C); e^* : $p < 0.01$ when compared to the number of tertiary follicles (TF) of control group (C); i : $p < 0.05$ when compared to the number of corpus luteum of control group (C); k : $p < 0.05$ when compared to the number of atretic follicles (FA) of control group (C); m : $p < 0.05$ when compared to the number of cystic follicles in control group (C)

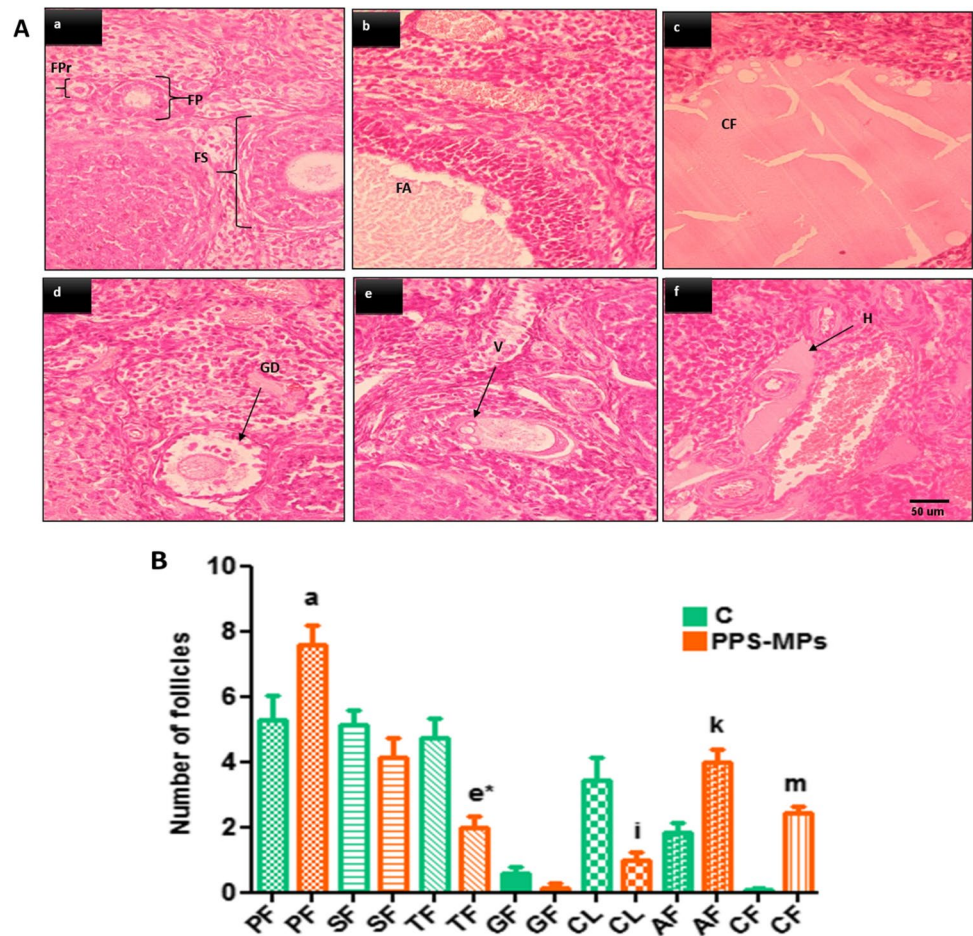


Table 2 Effect of polystyrene microplastic particles (PPS-MPs) on serum level of estradiol (E2) and on some markers of oxidative stress (CAT, catalase; SOD, superoxide dismutase; PSH, sulfuryl proteins; MDA, malondialdehyde)

Groups	Control (C)	Exposed group (PPS-MPs)
E2 (ng/ml)	28.78 ± 9.75	11.63 ± 7.13 ^a
CAT (μmol/min/mg of protein)	24.15 ± 6.34	47.75 ± 16.87 ^a
SOD (μmol/mg of protein)	42.17 ± 19.51	70.10 ± 4.34
PSH (μg/mg of protein)	5.78 ± 2.65	1.02 ± 0.23 ^a
MDA (μmol/mg of protein)	0.023 ± 0.10	0.052 ± 0.01 ^a

a: $p < 0.05$ when compared to the control group (C)

Effect on α -tubulin and DAAM-1 expression

To evaluate the effect of PPS-MP exposure on the cytoskeleton, we carried out an immunofluorescence assay of two cytoskeleton proteins: α -tubulin (DAAM-1) markers. The results in Figs. 5 and 6 show that PPS-MP exposure induced a significant decrease ($p < 0.05$) in the expression of α -tubulin and DAAM-1 proteins.

The relative expression of the 2 genes encoding α -tubulin and DAAM-1 was analyzed by RT-PCR. The results in Fig. 7 show that α -tubulin and DAAM-1 genes were significantly ($p < 0.01$) down-regulated of 0.52 ± 0.19 and 0.28 ± 0.14 -fold respectively compared to the controls.

Discussion

It has been shown that MPs have toxicological effects on different organisms, especially on aquatic animals. However, the assessment of the impact of these particles on mammalian animals, in which the results obtained can be transposed to humans, is limited. Furthermore, studies dealing with the effects of MPs on some parameters relating to the female reproductive function in mammalian models are almost absent. Thus, the main objective of our study was to assess the impact of PS-MP exposure on ovarian function in Wistar rats.

MPs with a diameter of 5 μ m are the smallest particles that can be ingested by aquatic organisms (Browne et al. 2008; Haemer et al. 2014; Watts et al. 2014). This, in turn, has increased concerns regarding human health exposure to MPs through the consumption of these contaminated organisms. Herein, we report the detection of FPS-MPs with a diameter of 5 μ m, as scattered fluorescent spheres, in the duodenum and, for the first time, in the different compartments of the ovarian tissue, following an experimental exposure of female rats during four estrous cycles. Using the same procedures, Deng et al. (2017) proved that 5 μ m of FPS-MPs accumulated in the duodenum, liver, and kidneys in the exposed mice during 28 days. The presence of PS-MPs in the ovarian tissue shows that these particles

Fig. 5 Immunofluorescence analysis of α -tubulin. **A** Representative photomicrographs of Control (a) and pristine polystyrene microplastic particle (PPS-MP)-exposed rats (b) ($Gr: \times 40$). **B** Histogram showing the quantification of α -tubulin fluorescence signal intensity using ImageJ. Values are expressed as means $\pm$ SE ($n = 7$ animals in each group). a: significant difference versus C group, $p < 0.05$

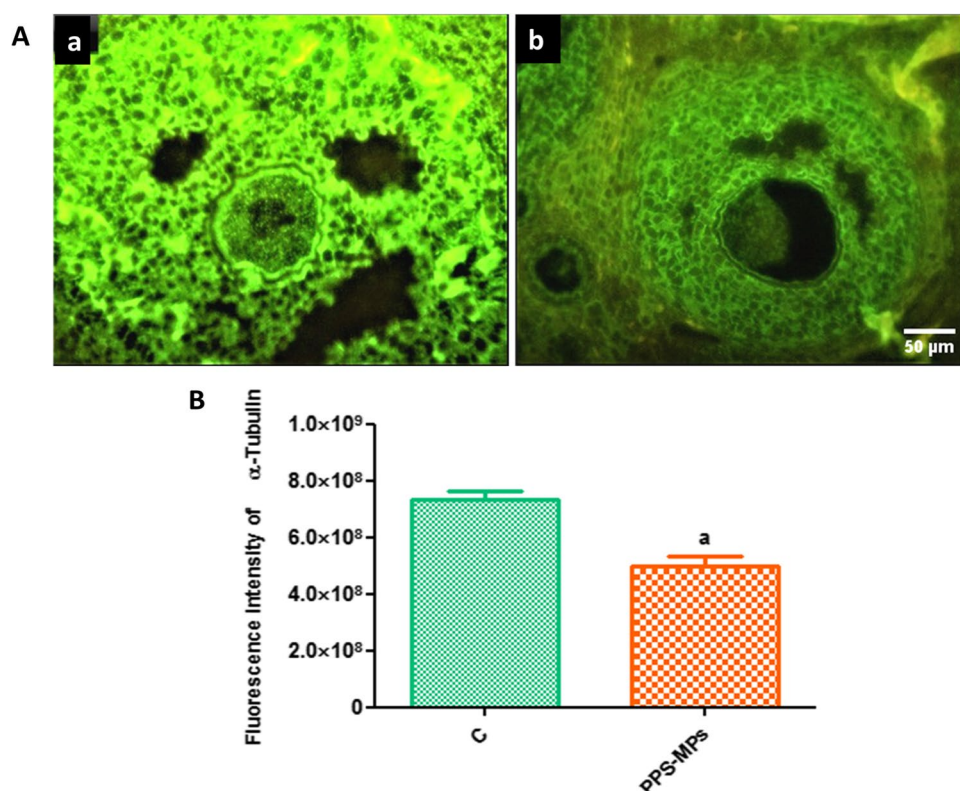


Fig. 6 Immunofluorescence analysis of DAAM-1. **A** Representative photomicrographs of ovaries from control (C) (**A a**) and pristine polystyrene microplastic particle (PPS-MP)-exposed rats (**A b**) ($Gr \times 40$). **B** Histogram showing the quantification of DAAM-1 fluorescence signal intensity using ImageJ. Values are expressed as means $\pm$ SE ($n=7$ animals in each group). a: significant difference versus C group, $p < 0.05$

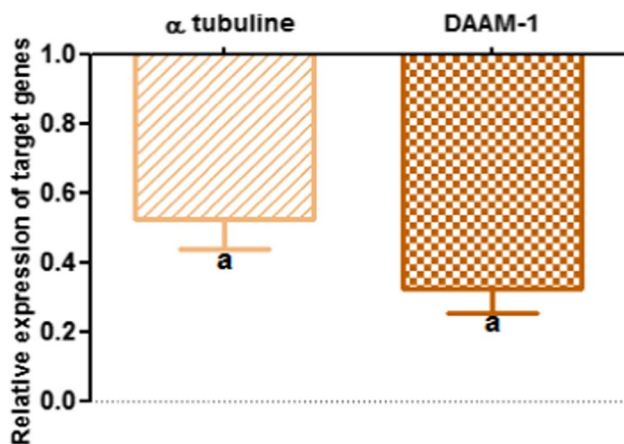
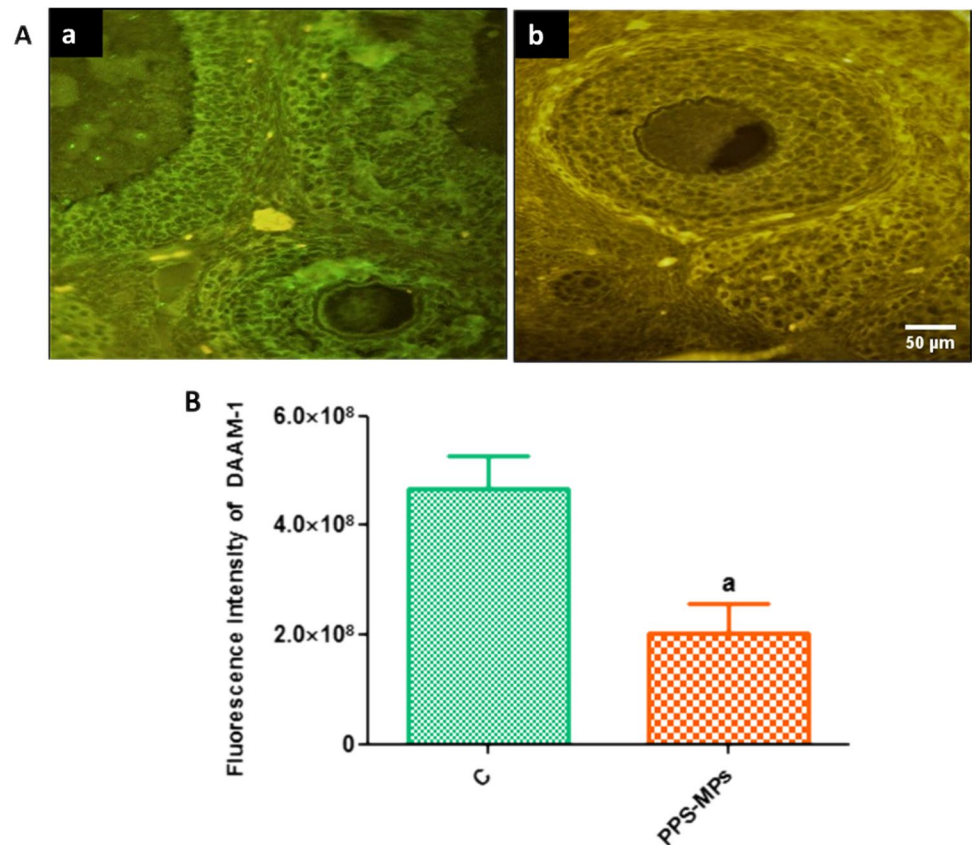


Fig. 7 Effects of pristine polystyrene microplastic particle (PPS-MP) exposure on ovarian gene expression of cytoskeleton markers (α -tubulin and DAAM-1) ($n=7$ animals in each group). a*: $p < 0.01$ when compared to control group

may have crossed the intestinal barrier and have been transported to the ovary by the circulatory system. Translocation of MPs at nanometer level from the intestine to the circulatory system and other tissues has also been described in several aquatic animals (Farrell and Nelson 2013; Triebkorn et al. 2019; Elizalde-Velázquez et al. 2020); however, results concerning the translocation of particles having the

size of micrometer remain inconclusive and contradictory. There are no convincing mechanisms presented in the literature to precisely explain this translocation of MPs in tissues. MPs of less than $10 \mu\text{m}$ could be translocated in the Peyer's patches of the ileum (the third portion of the small intestine) (Wright and Kelly 2017), or could enter the tissues via paracellular persorption (Volkheimer and Schulz 1998). Higher levels of accumulated PS-MPs in the luteal cells of the corpus luteum and in the thecal cells of the follicles, compared to other cells, can be explained by their contact with the blood capillaries. The accumulation of FPS-MPs in follicles, observed in our study, as well as in granulosa cells (An et al. 2021), indicates that PS-MPs have crossed not only the endothelial barrier of blood vessels, but also the blood-follicular barrier. The translocation of spherical polystyrene nanoparticles (25 and 50 nm) into the testis blood barrier has also been recently noted by Amereh et al. (2020). Thus, PS-MPs may have adverse effects on the reproductive system.

The present study shows that the toxicity of accumulated PS-MPs was manifested by the reduced body and relative ovarian weights, by the alteration in the estrous cycle duration, the altered histological structure, and by a reduction of serum E_2 concentration. The decrease in weight gain at the end of the exposure period is in accordance with many previous studies (Bruck and Ford 2018; Lu et al. 2018a; Luo

et al. 2019; Qiao et al. 2019). In most of these studies, it has been shown that the effect of MPs on body weight depends on the size and the concentration of MPs used and that relatively high concentrations do cause a decrease in body weight gain. In this context, Bruck and Ford (2018) suggested that the weight development of amphipods depended on the concentration of 8 μm MPs. Our suggestion is that MPs accumulated in the duodenum can cause mechanical damage to the digestive system, increasing energy expenditure, and even hormonal disturbances which lead to false satiety as described in mussels (Sussarellu et al. 2016) and zebrafish (Lu et al. 2016). The decrease in body weight gain observed in our study can also be explained, partly, by the change in the energetic metabolic profile. Accordingly, Lu et al. reported that the body weights began to decrease significantly at the third week in mice exposed to 0.5 μm PS-MPs and began to decrease significantly at the fourth week in those exposed to 50 μm PS-MPs (Lu et al. 2018a). In this study, it was observed that PS-MPs induced hepatic lipid metabolism disorder in mice. Besides, exposure of zebrafish larvae to 5 μm PS-MPs altered metabolites involved in energy and glycolipid metabolism (Wan et al. 2019).

In the current study, to our knowledge, we provide the first evidence that adult exposure to 5 μm PS-MPs leads to an abnormal estrous cycle and folliculogenesis. A significant decrease in the duration of the fourth estrus cycle of the treated group was found. The decrease in duration essentially affects the metestrus phase after 15 days of exposure to PS-MPs. Altered estrous cyclicity occurs together with a disruption of the maturation and differentiation of follicles as proved by a significant increase in the number of primary follicles and a significant decrease in the number of tertiary follicles and corpus luteum. These changes can be justified by a significant increase in the number of atretic and cysts follicles and can also lead to a rapid depletion of follicular stock. All the described ovarian alterations may be a possible consequence of an impaired steroid availability and production at both the central and local level. The decrease of the serum concentration of E_2 observed in PS-MP-treated animals may be a consequence of the effect of plastic particles on the hypothalamic-pituitary axis. In fact, one study has reported that PS-MPs interfere with the hypothalamic-pituitary axis and induce a change in FSH and LH concentrations (Amereh et al. 2020). Follicular atresia, and particularly the degeneration of granulosa cells, is probably a cause of E_2 depletion since these cells have the role of aromatizing androgens into E_2 . Accordingly, Amereh et al. (2020) showed that polystyrene nanoplastics lowered testosterone levels accompanied by a decrease in the number of Sertoli cells which aromatize androgens into testosterone. Since thecal and luteal cells are essential for the secretion of E_2 , we can note that the presence of PS-MPs in these two cells can be another cause of this hormonal imbalance. The alteration

of lipid metabolism resulting from exposure to 5 μm PS-MPs, described in some studies (Lu et al. 2016; Wan et al. 2019), could also explain the decrease of the serum concentration of E_2 observed in PS-MP-treated animals.

The affected ovarian function following PS-MP treatment might be the effect of the induction of oxidative stress, which was supported by the increased MDA concentration, a marker of the enhanced lipid peroxidation, the increased SOD and CAT activities as well as the decrease in PSH level, a marker of protein oxidation, in the rat ovary. The increase in SOD and CAT activities can be explained by the fact that (Nrf2), one of the components of the signaling pathway (Nrf2/ARE signaling) orchestrating the transcription of the *in vivo* antioxidant genes of SOD and CAT, increases at the protein level in the early stage of oxidative stress thereby cause the increase of SOD and CAT at the mRNA and protein levels (Jaiswal 2004). Accordingly, several studies proved that nanometric and micrometric PS-MP exposure induced a perturbation in oxidative status in mice testis (Amereh et al. 2020; Xie et al. 2020) and liver (Deng et al. 2017). Moreover, An et al. (2021) showed that reactive oxygen species (ROS) level was significantly elevated in granulosa cells treated *in vitro* by 0.5 μm PS-MPs and anti-oxidant N-Acetyl-L-cysteine inhibited the ROS level, suggesting that PS-MPs induced oxidative stress in ovary by increasing ROS production. Likewise, inducing oxidative stress was described as the main mechanism of PS-MP toxicity in several aquatic organisms (Jeong et al. 2017; Hamed et al. 2020; Zhu et al. 2020). However, it is not excluded that oxidative stress is a side effect of the immunotoxicity induced by MPs, since studies conducted mainly in invertebrates but also in vertebrates have demonstrated a compromised immune system under MP exposure (Hirt and Body-Malapel 2020).

The cytoskeleton is one of the most sensitive structures to xenobiotics and it is considered to be the primary target of ROS (Wilson and González-Billault 2015). This structure presents the internal architecture of cells and it is made up of fibrous networks that play an important role in transport and cell division. α -tubulin is the subunit of tubulin, a major component of the cytoskeleton (Goldman et al. 1979). During oocyte maturation, α -tubulin is shown to be involved in cytoskeleton dynamics and cell cycle control (Lu et al. 2018b). DAAM-1 is a protein belonging to the formin family, a large group of molecules involved in the organization of the cytoskeleton (Yasunaga et al. 2015). The protein is located in the plasma membrane and cytoplasmic vesicles (Kida et al. 2007), and is a key component of the non-canonical WNT signaling pathway (Habas et al. 2001), which participates in cell movements, such as convergent extension and planar cell polarity (Veeman et al. 2003). Non-canonical WNTs are expressed in granulosa cells and their inactivation leads to female infertility associated with increased follicular

atresia and decreased ovulation rates (Abedini et al. 2016). Recently, An et al. (2021) have indicated that Wnt/ β -catenin signaling pathway is activated in granulosa cells by 0.5 μ m PS-MP-induced oxidative stress. In the present study, a significant decrease in the expression of cytoskeletal proteins, α -tubulin and DAAM-1 in the ovary of rats exposed to PS-MPs, compared to control rats, was demonstrated by immunofluorescence. Importantly, these results were confirmed by the molecular study which revealed that PS-MPs down-regulated the expression of genes encoding α -tubulin and DAAM-1 proteins. In accordance with our results, Magni et al. (2019) showed, by mass spectrometry, that exposure to PS-MPs (1 and 10 μ m) caused a modulation of cytoskeletal proteins in zebra mussels *Dreissena polymorpha*. The effect of MPs on cytoskeletal proteins has not been studied in the ovaries. However, some studies conducted on plasticizers are available in the literature. Wang et al. (2016) indicated that porcine oocyte maturation was disrupted after bisphenol A treatment through disrupting cytoskeletal dynamics. In the same way, Kalo et al. (2019) noted the detrimental effect of phthalates on the gene expression of some cytoskeletal markers. Whether the expression changes of cytoskeleton-related genes identified here such as α -tubulin and DAAM-1 are specific to PS-MPs or apply more generally to MPs would require further studies. Besides, other studies should be conducted in order to assess the possible involvement of MP-induced changes in the gene expression pattern of cytoskeletal markers in female infertility.

Conclusion

Herein, the first evidence that adult exposure to 5 μ m PS-MPs leads to an abnormal estrous cycle and defective folliculogenesis in the rat was provided. One of the pivotal results is the detection of PS-MPs in the different compartments of the ovary and its negative effect on the expression of α -tubulin and DAAM-1 proteins. These findings can provide new data support for women's reproductive health risk that can be caused by MPs. All these results seem to us of great interest and other experiments should be conducted on the underlying mechanisms involved in PS-MP-induced change in the gene expression pattern of cytoskeletal markers and their impact on female infertility.

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Author contribution Conceptualization, Imed Messaoudi and Mohamed Banni; data curation, Asma Haddadi and Kaouthar Kessabi; formal analysis, Imed Messaoudi and Mohamed Banni; investigation, Asma Haddadi, Kaouthar Kessabi, Sana Boughammoura, Mariem

Ben Rhouma, and Rania Mlouka; methodology, Kaouthar Kessabi, Sana Boughammoura, and Mariem Ben Rhouma; project administration, Imed Messaoudi; resources, Imed Messaoudi; supervision, Imed Messaoudi and Mohamed Banni; validation, Kaouthar Kessabi and Rania Mlouka; visualization, Mohamed Banni; writing—original draft, Asma Haddadi and Kaouthar Kessabi; writing—review and editing, Mohamed Banni and Imed Messaoudi.

Data Availability All data generated or analyzed during this study are included in this published article.

Declarations

Ethics approval The experiments were approved by the local ethics committee of Institute of Biotechnology, University of Monastir, Ref: CER-SVS/ISBM022/2020.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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